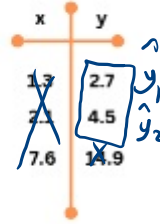
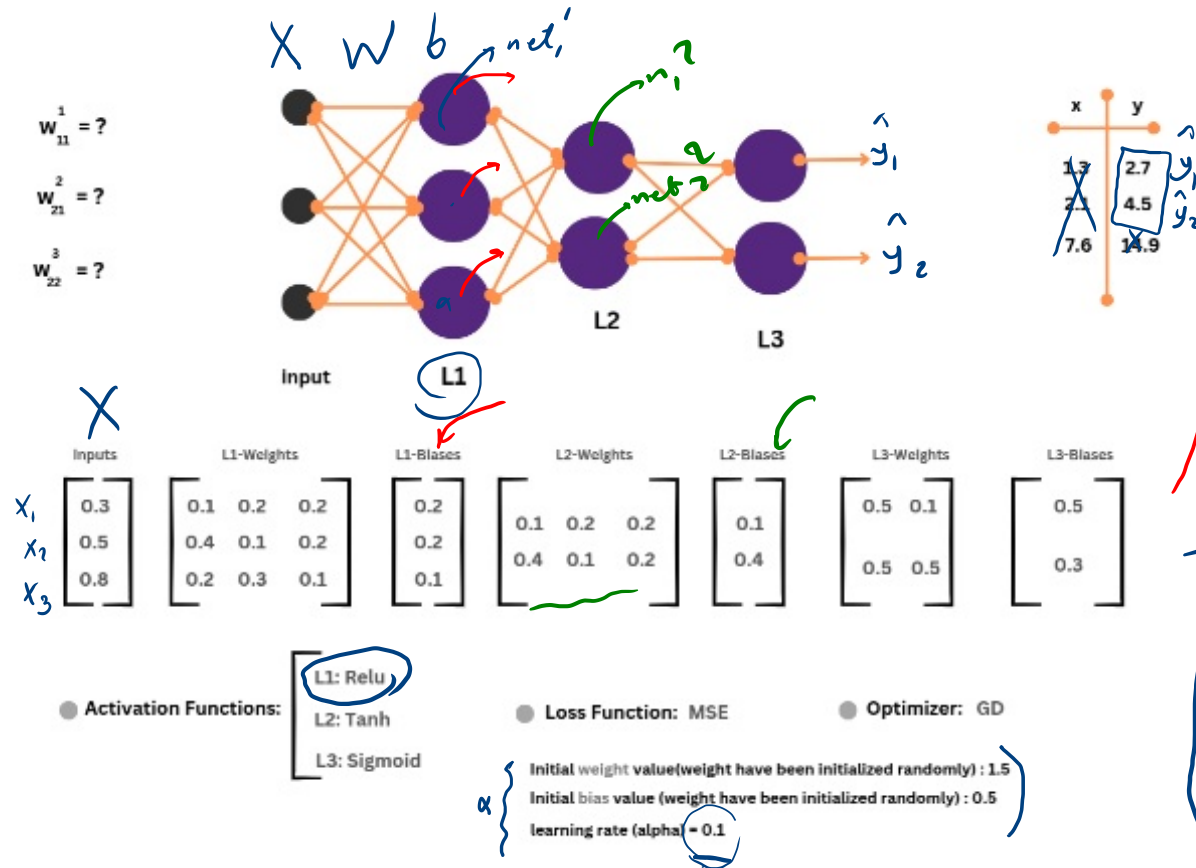


4. For the below feed-forward neural network architecture obtain the following weights after 1st Back Propagation iteration (epoch): [ignore the biases change during backpropagation]



Handwritten calculations for the first layer (L1):

$$net_1 = \begin{bmatrix} net_1' \\ net_2' \\ net_3' \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 4 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 7 \end{bmatrix}$$

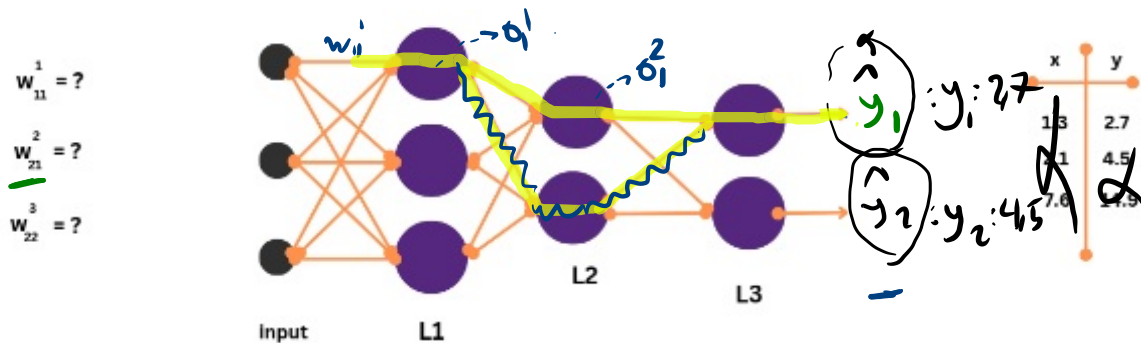
$$O_1 = \text{Relu} \begin{bmatrix} net_1' \\ net_2' \\ net_3' \end{bmatrix} = \begin{bmatrix} \text{Relu}(net_1') \\ \text{Relu}(net_2') \\ \text{Relu}(net_3') \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.5 \end{bmatrix}$$

Handwritten calculations for the second layer (L2):

$$w_2 O_1 + b_2 \Rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.2 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} net_1^2 \\ net_2^2 \end{bmatrix}$$

$$O = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} = \begin{pmatrix} \tanh(net_1^2) \\ \tanh(net_2^2) \end{pmatrix}$$

4. For the below feed-forward neural network architecture obtain the following weights after 1st Back Propagation iteration (epoch): [ignore the biases change during backpropagation]



x_1
 x_2
 x_3

Inputs	L1-Weights	L1-Biases	L2-Weights	L2-Biases	L3-Weights	L3-Biases
$\begin{bmatrix} 0.3 \\ 0.5 \\ 0.8 \end{bmatrix}$	$\begin{bmatrix} 0.1 & 0.2 & 0.2 \\ 0.4 & 0.1 & 0.2 \\ 0.2 & 0.3 & 0.1 \end{bmatrix}$	$\begin{bmatrix} 0.2 \\ 0.2 \\ 0.1 \end{bmatrix}$	$\begin{bmatrix} 0.1 & 0.2 & 0.2 \\ 0.4 & 0.1 & 0.2 \end{bmatrix}$	$\begin{bmatrix} 0.1 \\ 0.4 \end{bmatrix}$	$\begin{bmatrix} 0.5 & 0.1 \\ 0.5 & 0.5 \end{bmatrix}$	$\begin{bmatrix} 0.5 \\ 0.3 \end{bmatrix}$

● Activation Functions:
 L1: Relu
 L2: Tanh
 L3: Sigmoid

● Loss Function: MSE

● Optimizer: GD

α { Initial weight value (weight have been initialized randomly) : 1.5
 Initial bias value (weight have been initialized randomly) : 0.5
 learning rate (alpha) = 0.1

$$L_1 = \text{MSE}_1 = \frac{1}{2} \sum (y_1 - \hat{y}_1)^2 = (y_1 - \hat{y}_1)^2$$

$$L_2 = (y_2 - \hat{y}_2)^2$$

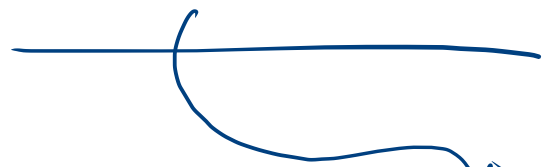
$$L = L_1 + L_2$$

$$w_{21}^{\text{new}} = w_{21}^{\text{old}} - \alpha \frac{\partial L}{\partial w_{21}} = w_{21}^{\text{old}} - \alpha \left(\frac{\partial L_1 + L_2}{\partial w_{21}} \right)$$

$$= w_{21}^{\text{old}} - \alpha \left[\frac{\partial L_1}{\partial w_{21}} + \frac{\partial L_2}{\partial w_{21}} \right] =$$

$$\frac{\partial L}{\partial w_{21}} = \frac{\partial L_1}{\partial o_1^3} \frac{\partial o_1^3}{\partial \text{net}_1^3} \frac{\partial \text{net}_1^3}{\partial o_1^2} \frac{\partial o_1^2}{\partial \text{net}_1^2} \frac{\partial \text{net}_1^2}{\partial o_1^1} \frac{\partial o_1^1}{\partial \text{net}_1^1} \frac{\partial \text{net}_1^1}{\partial w_{21}} + \dots$$

Forecast



Recurrent neural nets

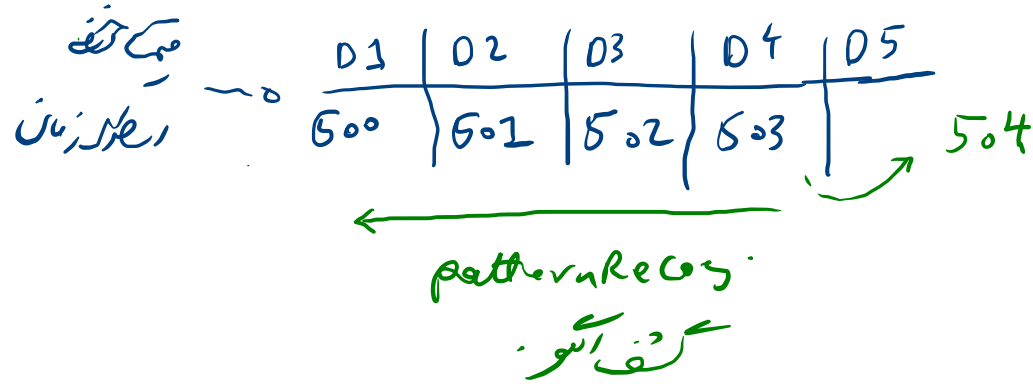
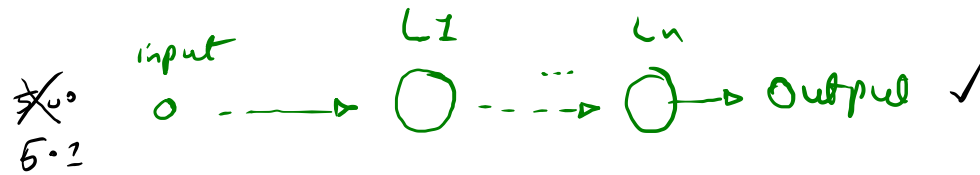
RNN

نیمه (رابطه) و RNN نیز به فرم سری استوخر می‌تواند (Time Series
Sequential Data)

باید دارای حافظه باشد

Dense

Feed Forward



House price forecasting → Tehran

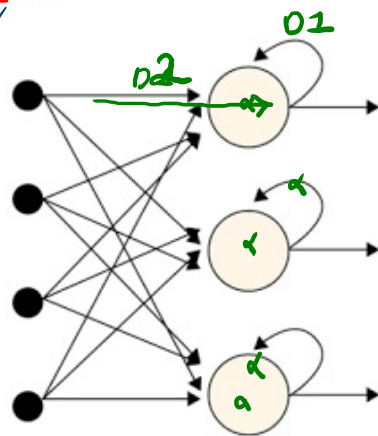
	area	floor	age	material	price	SD
D1	30	5	30	1	500	1
D2	"	"	"	"	503	1
Dn	"	"	"	"	590	1
						2

مؤلفه‌های ورودی به مدل
 در شبکه

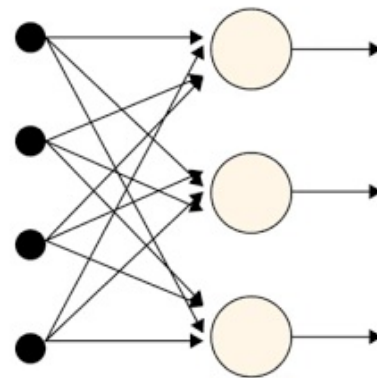
O_1
 O_2
 $\vdots$
 X_1
 X_2
 X_3
 X_4

3
 waro
 events
 price
 0,7
 foreign
 policy

(a) Recurrent Neural Network



(b) Feed-Forward Neural Network



SCALER
 Topics

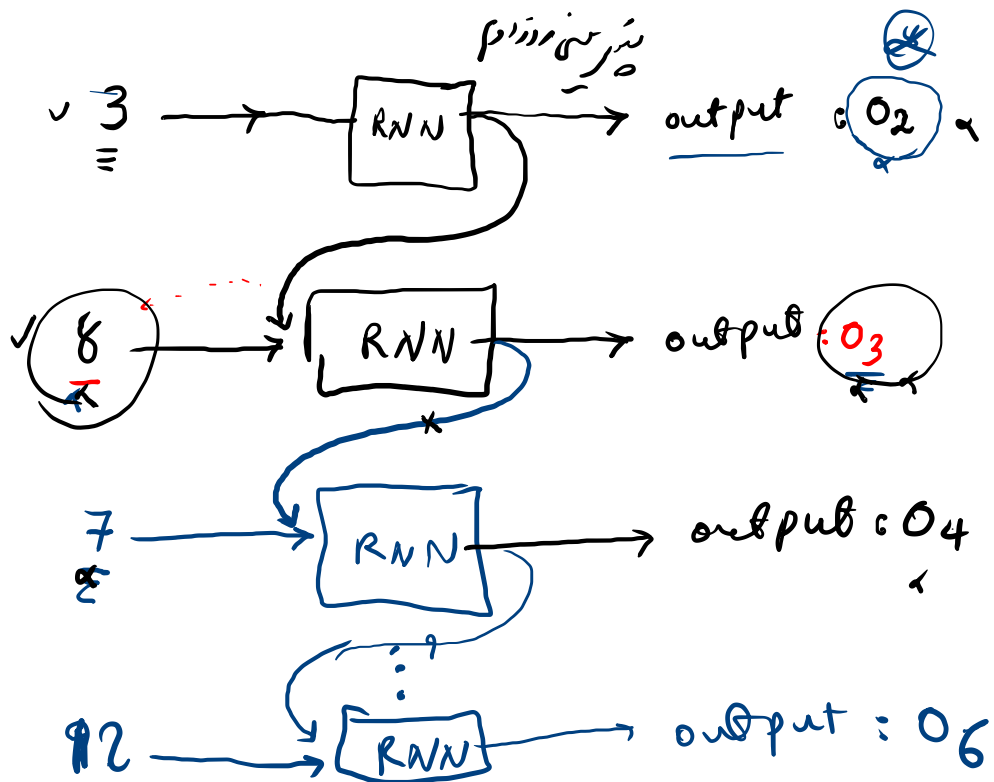
این یک
RNN

[Maine RNN]

t

factor

1	3
2	8
3	7
4	9
5	12
6	?



$L(O_2, 8) : \min$

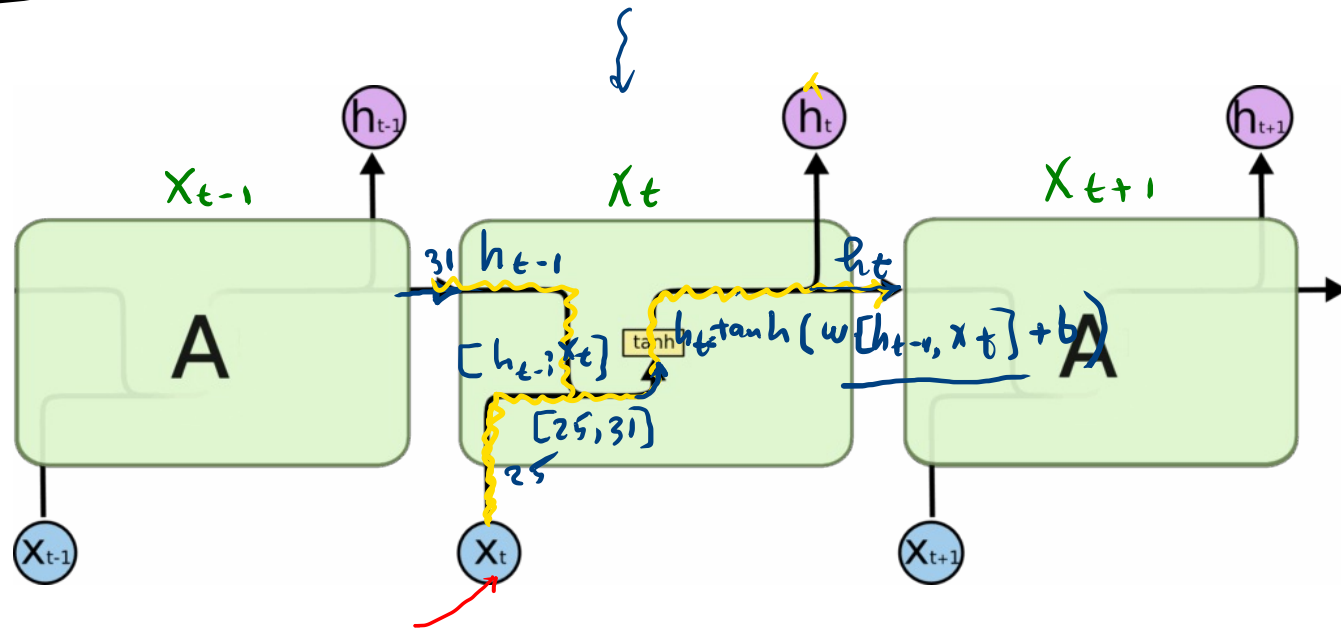
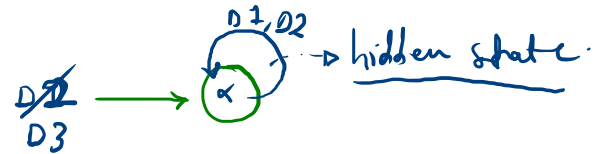
روزگار ← $L(O_3, 7) : \min$

...

basic
vanilla RNN

Price

D1	D2	D3
3	6	9



1	2
2	<u>3</u>
3	<u>4</u>

$$h_{t-1} = 0$$

$$\underline{w, b = 1}$$

$$h_1 = \tanh (w [\underline{0}, \underline{2}] + b) = \tanh ([0, 2] + 1) = [\tanh (0+1), \tanh (2+1)]$$

$$h_2 = \tanh (w (\overset{h_1}{\underline{a}}, \underline{3}) + b) = \tanh (\underline{1}, \underline{3}) = \underbrace{[\tanh (1), \tanh (3)]}_a$$

$$h_3 = \tanh (w (h_2, \underline{4}) + b)$$

⋮

RNN; weight sharing

meta p₄ → 7

	10	9	8	7	6	5	4	3	2	1
meta p ₄	640	643	640	653	655	653	650	681	680	688

	D1	D2	D3	D4	D5	D6	D7	forecast next Day
X _{t-2}	640	643	640	653	655	653	650	681
X _{t-1}	643	640	653	653	653	650	681	680
X _t	640	653	655	653	650	681	680	688
X _{t+1}								

meta [طراحی عمل، پارامتر]

پارامترهای متی و پارامتر

در روز آینده! با توجه به تاریخچه پیش

$$L(h_{t-2}, 681)$$

$$h_{t-2} = \tanh(w[o, x_{t-2}] + b)$$

$$h_{t-1} = \tanh(w[h_{t-2}, x_{t-1}] + b)$$

$$h_t = \tanh(w(h_{t-1}, x_t) + b)$$

after trans step → trans of w, b

$$x_{t+1} = \tanh(w(h_t, x_t) + b)$$

ویرایش

mlp

Temp

N-gram

[———] [/ / /] [/ / /] [/ / /] [/ / /]
yesterday was so hot, but

before that was so [———]
[/ / / /] [/ / /] [/ / /] [/ / /]
a

